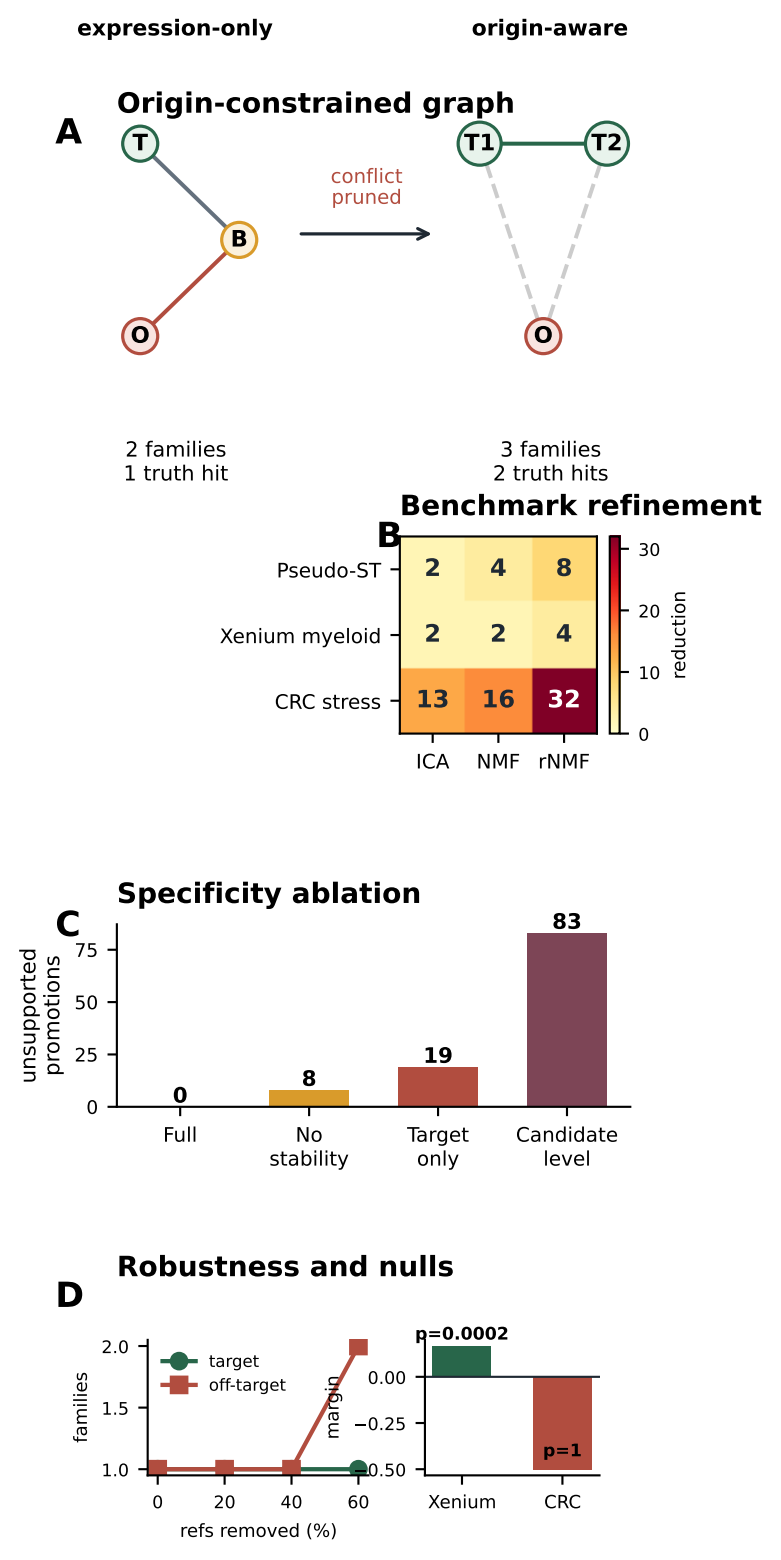
expression-only
origin-aware
A
Origin-constrained graph
T
B
O
conflict
pruned
T1
T2
O
2 families
1 truth hit
3 families
2 truth hits
B
Benchmark refinement
Pseudo-ST
Xenium myeloid
CRC stress
2
4
8
2
2
4
13
16
32
ICA
NMF
rNMF
30
20
10
0
reduction
C
Specificity ablation
unsupported
promotions
75
50
25
0
0
8
19
83
Full
No
stability
Target
only
Candidate
level
D
Robustness and nulls
families
2.0
1.5
1.0
target
off-target
0
20
40
60
refs removed (%)
margin
p=0.0002
0.00
−0.25
−0.50
p=1
Xenium
CRC